

FRANKFURT HAHN, Germany

WMO: 106160

Lat: 49.9461N

Lon: 7.2644E

Elev: 1651

StdP: 13.84

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	19.3	8.7	9.1	21.4	13.6	11.7	21.7	28.0	45.1	25.4	42.3	8.3	140	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	82.8	65.4	78.9	63.8	75.5	62.5	67.1	78.5	65.5	75.9	64.0	72.8	8.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	93.3	70.2	62.3	89.1	68.8	60.7	84.2	67.4	32.5	78.5	31.2	76.0	30.1	72.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.4	19.6	17.4	DB	11.2	89.0	5.3	3.7	7.4	91.6	4.3	93.7	1.4	95.8	-2.5	98.5
			WB	10.4	70.9	5.3	1.6	6.6	72.1	3.5	73.0	0.5	73.9	-3.3	75.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	32.8	34.0	40.3	47.6	54.4	60.2	63.4	62.9	56.4	48.5	39.9	34.7 (6)
	DBStd	12.97	7.67	7.51	7.34	7.52	6.87	6.60	6.50	6.20	5.90	6.57	6.57	7.10 (7)
	HDD50	2357	535	447	308	132	34	3	0	0	9	107	306	476 (8)
	HDD65	6355	1000	867	766	521	334	173	110	114	266	512	752	940 (9)
	CDD50	1631	0	0	7	61	171	308	416	401	201	61	5	0 (10)
	CDD65	153	0	0	0	0	5	29	61	50	8	0	0	0 (11)
	CDH74	1105	0	0	0	2	34	206	433	388	42	0	0	0 (12)
	CDH80	273	0	0	0	0	3	44	104	120	2	0	0	0 (13)
(14) Wind	WSAvg	8.7	10.2	10.0	9.4	8.3	7.9	7.4	7.5	7.3	7.8	9.1	9.2	9.9 (14)
(15) Precipitation	PrecAvg	28.0	2.4	2.2	2.0	1.7	2.4	2.5	2.6	2.6	2.1	2.3	2.4	2.7 (15)
	PrecMax	38.5	7.8	6.3	5.4	3.9	5.1	5.6	7.7	5.2	4.8	7.2	4.6	8.1 (16)
	PrecMin	21.2	0.3	0.2	0.2	0.0	0.6	0.7	1.1	0.6	0.6	0.4	0.1	0.2 (17)
	PrecStd	4.2	1.4	1.4	1.3	0.9	1.1	1.2	1.4	1.2	1.1	1.3	1.1	1.5 (18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	53.9	62.7	72.6	78.5	85.6	87.9	89.7	79.0	69.7	58.6	51.9 (19)
		MCWB	48.5	45.4	49.5	55.5	63.7	68.1	66.3	66.3	62.4	59.6	54.3	49.1 (20)
	2%	DB	48.6	49.4	58.2	68.2	73.2	79.7	82.7	82.8	73.8	64.3	54.2	49.7 (21)
		MCWB	46.6	44.5	47.5	53.5	60.1	64.5	65.1	65.4	61.5	57.6	51.0	47.6 (22)
	5%	DB	46.3	46.6	53.8	64.1	69.6	75.3	78.8	77.5	69.8	60.6	51.7	46.8 (23)
		MCWB	44.6	43.9	46.2	52.2	57.8	63.2	64.0	63.2	59.9	56.4	49.4	45.5 (24)
	10%	DB	43.0	44.5	50.4	59.9	66.2	71.7	75.2	73.7	66.2	57.4	48.6	44.5 (25)
		MCWB	41.7	42.2	45.1	50.5	56.3	61.4	62.2	62.1	58.6	54.3	47.0	43.5 (26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.7	47.7	51.3	58.0	65.7	69.5	69.2	68.7	65.2	60.9	55.2	50.2 (27)
		MCDB	50.3	50.7	57.5	68.5	74.8	80.2	81.8	81.5	73.8	66.2	57.5	51.1 (28)
	2%	WB	47.0	45.8	49.5	55.2	62.1	66.4	66.9	66.6	63.0	58.4	51.8	47.8 (29)
		MCDB	48.3	47.9	54.7	65.3	70.3	76.6	78.3	78.6	70.8	62.4	53.9	49.0 (30)
	5%	WB	44.5	44.1	47.8	53.1	59.6	64.3	65.3	65.0	61.2	56.8	49.6	45.6 (31)
		MCDB	46.1	46.3	52.4	61.5	67.1	73.0	75.8	75.0	68.0	60.2	51.6	46.8 (32)
	10%	WB	41.7	42.0	46.0	51.1	57.3	62.5	63.7	63.2	59.3	54.7	47.3	43.2 (33)
		MCDB	42.9	44.2	50.1	58.9	63.9	70.1	72.7	71.6	64.9	57.5	48.7	44.1 (34)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.6	11.8	14.6	14.8	15.7	15.4	15.2	13.2	10.6	7.6	7.0 (35)
		MCDBR	9.0	12.2	17.2	20.1	19.2	20.6	21.0	21.6	18.5	15.0	11.0	9.0 (36)
	5% WB	MCWBR	8.4	8.9	10.0	10.3	10.1	10.6	9.2	9.8	9.8	9.9	8.7	8.0 (37)
		MCWBR	8.8	10.2	14.2	18.1	17.3	18.4	18.3	18.2	16.0	13.8	10.1	8.5 (38)
(40) Clear-Sky Solar Irradiance	taub		0.301	0.327	0.357	0.383	0.390	0.392	0.391	0.387	0.361	0.348	0.322	0.299 (40)
		taud	2.438	2.388	2.322	2.289	2.295	2.307	2.299	2.340	2.395	2.426	2.436	2.427 (41)
	Ebn at Noon		238	258	268	273	275	275	273	268	264	246	225	220 (42)
		Edh at Noon	20	26	33	38	40	39	39	36	30	25	20	17 (43)
(44) All-Sky Solar Radiation	RadAvg		250	477	847	1335	1574	1747	1656	1419	1061	613	283	194 (44)
	RadStd		38	90	129	164	158	159	180	144	129	59	34	29 (45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page